Data Log

Set Up

- Imported 2019_Q2, 2019_Q3, 2019_Q4, and 2020_Q1 into schema cyclistic_case_study
- Created backup tables in schema cyclistic_case_study_backups
- Normalized column names. Table at end of log.
- Normalized usertype values based on report vocabulary. Member/Casual

```
--find usertype names
select usertype
from cyclistic_case_study."2019_Q3"
group by usertype;
--change usertype term for Member
update cyclistic_case_study."2020_Q1"
set usertype = replace(usertype, 'Subscriber', 'Member');
--change usertype term for Casual
update cyclistic_case_study."2020_Q1"
set usertype = replace(usertype, 'customer', 'Casual');
```

- Added trip_duration to "2020_Q1"

```
alter table cyclistic_case_study."2020_Q1"
add column trip_duration int4
```

- Computed and checked trip_duration for "2020_Q1"

```
--subtract start_time from end time to get seconds of trip
update cyclistic_case_study."2020_Q1"
set trip_duration = extract(epoch from end_time - start_time);
--show 5 values to check work
select end_time, start_time, trip_duration
from cyclistic_case_study."2020_Q1"
limit 5;
```

- Added null columns for gender and birthyear to "2020_Q1"

```
alter table cyclistic_case_study."2020_Q1"
add column birthyear int4,
add column gender varchar(50)
```

- Checked that all columns existed and data type was correct.

```
--Claude Sonnet 4.6 Assisted
select table_name, column_name, data_type
from information_schema.columns
where table_schema = 'cyclistic_case_study'
and table_name in ('2019_Q2', '2019_Q3', '2019_Q4', '2020_Q1')
and column_name in ('trip_id', 'start_time', 'end_time', 'start_station_name',
'start_station_id', 'end_station_name', 'end_station_id','usertype',
'trip_duration', 'gender', 'birthyear')
order by column_name, table_name;
```

- Aggregated trip_id, start_time, end_time, start_station_name_start_station_id, end_station_name, end_station_id, usertype, trip_duration, gender, and birthyear for all 4 quarters into single table, four_quarters

```
--Claude Sonnet 4.6 Assisted
create table cyclistic_case_study.four_quarters as

select trip_id, start_time, end_time, start_station_name, start_station_id,
end_station_name, end_station_id, usertype, trip_duration, gender, birthyear
from cyclistic_case_study."2019_Q2"
union all

select trip_id, start_time, end_time, start_station_name, start_station_id,
end_station_name, end_station_id, usertype, trip_duration, gender, birthyear
from cyclistic_case_study."2019_Q3"
union all

select trip_id, start_time, end_time, start_station_name, start_station_id,
end_station_name, end_station_id, usertype, trip_duration, gender, birthyear
from cyclistic_case_study."2019_Q4"
union all

select trip_id, start_time, end_time, start_station_name, start_station_id,
end_station_name, end_station_id, usertype, trip_duration, gender, birthyear
from cyclistic_case_study."2020_Q1";
```

- Added day_of_week column

```
--select extract(dow from start_time) as day,
alter table cyclistic_case_study.four_quarters
add column day_of_week varchar(50);

--Claude check before executing. "FMDay removes trailing spaces added by
Postgres."
update cyclistic_case_study.four_quarters
set day_of_week = to_char(start_time, 'FMDay');

--Result Check. look right? yes. okay good.
select *
from cyclistic_case_study.four_quarters fq
limit 20;
```

- Added age_demo column

```
alter table cyclistic_case_study.four_quarters
add column age_demo varchar(50);

--Claude Sonnet 4.6 Assisted. Find age and put into category, null for below 17 &
below 0.
update cyclistic_case_study.four_quarters
set age_demo = case
    when (2019 - birthyear) between 18 and 24 then '18-24'
    when (2019 - birthyear) between 25 and 34 then '25-34'
    when (2019 - birthyear) between 35 and 44 then '35-44'
    when (2019 - birthyear) between 45 and 54 then '45-54'
    when (2019 - birthyear) between 55 and 64 then '55-64'
    when (2019 - birthyear) >= 65 then '65+'
```

```
    else null
end;

--Result Check. look right? yes. okay good.
select birthyear, age_demo, (2019-birthyear) as calculated_age
from cyclistic_case_study.four_quarters fq
where birthyear is not null
limit 20
```

- Asked Claude what else might be useful to calculate. Recommendations: ~~month~~, ~~season~~, ~~hour_of_day~~, time_of_day, ~~is_weekend~~, ~~is_round_trip~~, trip_duration_bucket, ~~start_station_frequency~~
- Created time_of_day column. *Will consider trip_duration_bucket after init analysis of trip_duration*

```
--Claude Sonnet 4.6 Assisted. Might reconsider time categories to add early morning.
alter table cyclistic_case_study.four_quarters
add column time_of_day VARCHAR(20);

update cyclistic_case_study.four_quarters
set time_of_day = case
    when extract(hour from start_time) >= 5  and extract(hour from start_time) <12 then 'Morning'
    when extract(hour from start_time) >= 12 and extract(hour from start_time) < 17 then 'Afternoon'
    when extract(hour from start_time) >= 17 and extract(hour from start_time) < 21 then 'Evening'
    else 'Night'
end;
```

Data Validation

Checked number of casual vs member riders from four_quarters and unformatted datasets

```
--Claude Sonnet 4.6 Assisted. This melted my brain. One day. One day.
SELECT 'backup' AS source,
    CASE
        WHEN usertype IN ('Subscriber', 'member') THEN 'Member'
        WHEN usertype IN ('Customer', 'customer', 'casual') THEN 'Casual'
    END AS usertype,
    COUNT(*) AS count
FROM (
    SELECT "User Type" AS usertype FROM cyclistic_case_study_backups."2019_Q2 (Backup)"
    UNION ALL
    SELECT usertype FROM cyclistic_case_study_backups."2019_Q3 (Backup)"
    UNION ALL
    SELECT usertype FROM cyclistic_case_study_backups."2019_Q4 (Backup)"
    UNION ALL
    SELECT member_casual AS usertype FROM cyclistic_case_study_backups."2020_Q1 (Backup)"
) AS backup_data
GROUP BY
    CASE
        WHEN usertype IN ('Subscriber', 'member') THEN 'Member'
        WHEN usertype IN ('Customer', 'customer', 'casual') THEN 'Casual'
```

```
    END

UNION ALL

SELECT 'four_quarters' AS source, usertype, COUNT(*) AS count
FROM cyclistic_case_study.four_quarters
GROUP BY usertype

ORDER BY usertype, source;
```

Output:

Source	Usertype	count
backup	Casual	1291454
four_quarters	Casual	1291454
backup	Member	3525032
four_quarters	Member	3525032

Checked if values are consistent. (No Male from one set and male from another.)

```
select gender, count(*)
from cyclistic_case_study.four_quarters fq
group by gender
```

Output:

Gender	count
	760254
Female	1036071
Male	2593274
[NULL]	426887

[Null] is from 2020_Q1. “ ” is because gender is an option field. Need to change “ “ to NULL.

```
UPDATE cyclistic_case_study.four_quarters
SET gender = NULL
WHERE gender = '';
```

New Output:

Gender	count
Female	1036071
Male	2593274
[NULL]	1187141

Birthyear. Checking for ‘ ‘ vs NULL

```
select birthyear, count(*) as count
from cyclistic_case_study.four_quarters fq
group by birthyear
order by count desc
```

Output:

Birthyear	count
[NULL]	1164028
1992	236288

Noticed likely inaccurate birthyear dates

```
select birthyear, count(*) as count
from cyclistic_case_study.four_quarters fq
```

```
where birthyear < 1919
group by birthyear
order by birthyear
```

Output:

birthyear	count
1759	1
1790	1
1888	40
1889	6
1890	2
1899	170
1900	270
1901	19
1904	286
1905	6
1909	11
1910	6
1911	1
1912	2
1918	19

840 riders older than 100! Medical technology has come a long way.

birthyear	count
2002	2871
2003	1105
2014	5

3,976 riders under 18, not counting the 5 5-year-olds.

Revise age bracket to change birthyears before 1919 to NULL and include 16+ to lowest age bracket.

```
update cyclistic_case_study.four_quarters
set birthyear = case
       when birthyear < (2019-100) then null
       else birthyear
end;

update cyclistic_case_study.four_quarters
set age_demo = case
    when (2019 - birthyear) between 16 and 24 then '16-24'
    when (2019 - birthyear) between 25 and 34 then '25-34'
    when (2019 - birthyear) between 35 and 44 then '35-44'
    when (2019 - birthyear) between 45 and 54 then '45-54'
    when (2019 - birthyear) between 55 and 64 then '55-64'
    when (2019 - birthyear) >= 65 then '65+'
    else null
end;
```

Checking age_demo

```
select age_demo, count(*)
from cyclistic_case_study.four_quarters fq
group by age_demo
```

age_demo	count
16-24	437627
25-34	1883888
35-44	708863
45-54	360474
55-64	218839
65+	41922
[NULL]	1164873

Baseline Analysis

Full Report

```
select var, count(*), round(count(*) * 100.0 / sum(count(*)) over (), 2) as
"%"
from cyclistic_case_study.four_quarters
group by var
```

usertype	count	%
Casual	1291454	26.81
Member	3525032	73.19

gender	Count	%
Female	1036071	21.51
Male	2593274	53.84
[NULL]	1187141	24.65

age_demo	count	%
16-24	437627	9.09
25-34	1883888	39.11
35-44	708863	14.72
45-54	360474	7.48
55-64	218839	4.54
65+	41922	0.87
[NULL]	1164873	24.19

day_of_week	count	%
Friday	710845	14.76
Monday	703649	14.61
Saturday	663455	13.77
Sunday	557862	11.58
Thursday	727844	15.11
Tuesday	715586	14.86
Wednesday	737245	15.31

time_of_day	count	%
Afternoon	1571001	32.62

Evening	1455317	30.22
Morning	1428834	29.67
Night	361334	7.50

Usertype Contrast

```
--Claude assist to find partition by to separate % pools by usertype.
select usertype, var, count(*), round(count(*) * 100.0 / sum(count(*)) over
(partition by usertype), 2) as "%"
from cyclistic_case_study.four_quarters fq
group by usertype, var
```

usertype	gender	count	%
Casual	Female	198598	15.38
Casual	Male	309107	23.93
Casual	[NULL]	783749	60.69
Member	Female	837473	23.76
Member	Male	2284167	64.80
Member	[NULL]	403392	11.44

Usertype	age_demo	count	%
Casual	16-24	131487	10.18
Casual	25-34	253254	19.61
Casual	35-44	70648	5.47
Casual	45-54	38569	2.99
Casual	55-64	17469	1.35
Casual	65+	2052	0.16
Casual	[NULL]	777975	60.24
Member	16-24	306140	8.68
Member	25-34	1630634	46.26
Member	35-44	638215	18.11
Member	45-54	321905	9.13
Member	55-64	201370	5.71
Member	65+	39870	1.13
Member	[NULL]	386898	10.98

usertype	day_of_week	count	%
Casual	Friday	178285	13.80
Casual	Monday	150892	11.68
Casual	Saturday	305387	23.65
Casual	Sunday	241714	18.72
Casual	Thursday	152616	11.82
Casual	Tuesday	127470	9.87
Casual	Wednesday	135090	10.46
Member	Friday	532560	15.11
Member	Monday	552757	15.68
Member	Saturday	358068	10.16
Member	Sunday	316148	8.97
Member	Thursday	575228	16.32
Member	Tuesday	588116	16.68
Member	Wednesday	602155	17.08

Usertype	time_of_day	count	%
Casual	Afternoon	556060	43.06
Casual	Evening	359807	27.86
Casual	Morning	252815	19.58
Casual	Night	122772	9.51
Member	Afternoon	1014941	28.79
Member	Evening	1095510	31.08
Member	Morning	1176019	33.36
Member	Night	238562	6.77

Trip Duration:

```
select round(avg(trip_duration)/60, 2) as average_trip_duration
from cyclistic_case_study.four_quarters
```

Usertype	average_trip_duration
All	26.83
Casual	59.33
Member	14.92

2019_Q2	2019_Q3	2019_Q4	2020_Q1	Standarized Name
01 - Rental Details Rental ID *int4*	trip_id *int4*	trip_id *int4*	ride_id (varchar(50))	trip_id (varchar(50))
			rideable_type	
01 - Rental Details Local Start Time	start_time	start_time	started_at	start_time
01 - Rental Details Local End Time	end_time	end_time	ended_at	end_time
03 - Rental Start Station Name	from_station_name	from_station_name	start_station_name	start_station_name
03 - Rental Start Station ID	from_station_id	from_station_id	start_station_id	start_station_id
03 - Rental End Station Name	to_station_name	to_station_name	end_station_name	end_station_name
03 - Rental End Station ID	to_station_id	to_station_id	end_station_id	end_station_id
			start_lat	
			start_lng	
			end_lat	
			end_lng	
User Type	usertype	usertype	member_casual	usertype
01 - Rental Details Duration In Seconds Uncapped	tripduration	tripduration	ride_duration	trip_duration
Member Gender	gender	gender	Null	gender
05 - Member Details Member Birthday Year	birthyear	birthyear	null	birthyear
02 - Rental Details Bike ID	bikeid	bikeid		bikeid